Tromino Tiling: Mathematical Induction and Recursion

1 Tiling a Deficient Chessboard with Trominoes

A particularly elegant application of mathematical induction arises in a tiling problem. A *tromino* is a tile made from three unit squares. We shall use an L-shaped tromino, as shown in Figure **??**. Rotations and reflections of the tile are also allowed.

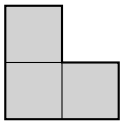


Figure 1: An L-shaped tromino consisting of three unit squares.

1.1 The Tromino Tiling Problem

Suppose one square is removed from a $2^n \times 2^n$ chessboard, where $n \geq 1$. Can the remaining board always be tiled completely using L-shaped trominoes, irrespective of which square is removed?

The answer is yes.

Let $P(n)$ denote the statement

$P(n)$: Every $2^n \times 2^n$ board with one square removed can be tiled with L-shaped trominoes.

We shall prove $P(n)$ by mathematical induction.

1.2 Proof by Mathematical Induction

Base case. For $n = 1$, the board has size

$$2^1 \times 2^1 = 2 \times 2.$$

It therefore consists of four unit squares. If any one square is removed, the remaining three squares form exactly one L-shaped tromino. Hence $P(1)$ is true.

Induction hypothesis. Suppose that $P(n)$ is true for some $n \geq 1$. Thus we assume that *any* $2^n \times 2^n$ board with one square removed can be tiled completely using L-shaped trominoes.

Inductive step. We must show that $P(n+1)$ is true. Consider a

$$2^{n+1} \times 2^{n+1}$$

board from which one arbitrary square has been removed.

Divide the board into four equal quadrants. Each quadrant is a $2^n \times 2^n$ board. The missing square must lie in exactly one of these four quadrants.

Without loss of generality, suppose that it lies in the top-left quadrant, as illustrated in Figure **??**.

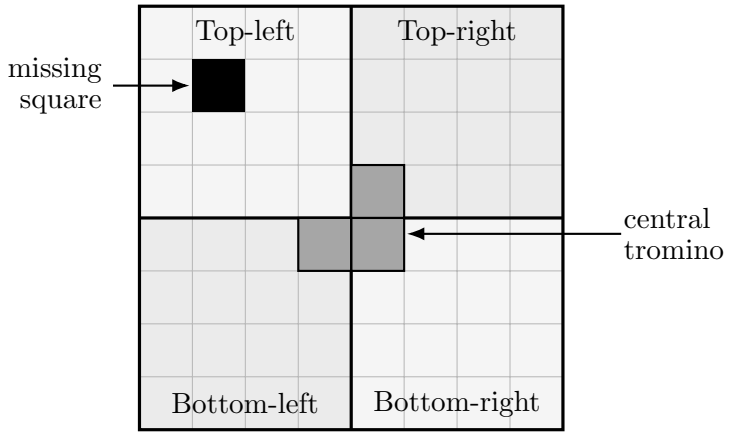


Figure 2: The $2^{n+1} \times 2^{n+1}$ board is divided into four $2^n \times 2^n$ quadrants. The original missing square is shown in black in the top-left quadrant. The shaded central L-shaped tromino occupies one square in each of the other three quadrants.

Now comes the crucial observation. The top-left quadrant already has one square missing. Place a single L-shaped tromino at the centre of the board so that it occupies one central square in *each of the other three quadrants.*

After placing this central tromino:

1. the top-left quadrant has exactly one square missing—the square that was originally removed;
2. the top-right quadrant has exactly one square unavailable—the square occupied by the central tromino;
3. the bottom-left quadrant similarly has exactly one square unavailable; and
4. the bottom-right quadrant also has exactly one square unavailable.

Thus we have transformed the original problem into four smaller problems of precisely the form covered by the induction hypothesis. Each quadrant is a $2^n \times 2^n$ board with one square missing.

By the induction hypothesis, each of the four quadrants can therefore be tiled completely with L-shaped trominoes. Together with the one tromino that we have already placed at the centre, these tilings cover the entire $2^{n+1} \times 2^{n+1}$ board except for the originally removed square.

Hence

$$P(n) \Longrightarrow P(n+1).$$

Since $P(1)$ is true, the Principle of Mathematical Induction gives

Every $2^n \times 2^n$ board with one square removed can be tiled completely with L-shaped trominoes.

2 The Inductive Proof is a Recursive Algorithm

The proof above does more than establish that a tiling exists. It actually tells us *how to construct one*. Hidden inside the inductive proof is therefore a recursive algorithm.

Suppose we are given a $2^n \times 2^n$ board and the position of its one missing square. Let us call the procedure

$$\textsc{Tile}(B, n, x),$$

where B is the board and x is its missing square.

The inductive proof suggests the following procedure.

1. If $n = 1$, then B is a 2×2 board. Since one square is missing, place one L-shaped tromino on the remaining three squares. The problem is solved.
2. Otherwise, divide B into four equal $2^{n-1} \times 2^{n-1}$ quadrants.
3. Determine which quadrant contains the missing square x.
4. Place one L-shaped tromino at the centre of the board so that it occupies one central square in each of the *other three* quadrants.
5. We now have four smaller boards, each with exactly one square missing:
   - in one quadrant it is the original missing square x;
   - in each of the other three quadrants it is the square occupied by the central tromino.
6. Apply exactly the same procedure recursively to each of these four smaller boards.

Thus a problem on a $2^n \times 2^n$ board is reduced to four problems on boards of size

$$2^{n-1} \times 2^{n-1}.$$

Each of these is again reduced to four still smaller problems, and so on, until we reach 2×2 boards.

2.1 The Recursive Algorithm

The construction can be written as the following algorithm.

2.2 The Recursion Tree

The recursive structure can be represented as a tree. A problem on a $2^n \times 2^n$ board generates four problems on $2^{n-1} \times 2^{n-1}$ boards. Each of those generates four still smaller problems.

Algorithm 1 Recursive algorithm for tiling a deficient $2^n \times 2^n$ board

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1: procedure TILE(B, n, x)
2:     if n = 1 then
3:         Place one L-shaped tromino on the three squares of B other than x
4:         return
5:     end if
6:     Divide B into four equal quadrants B_1, B_2, B_3, B_4
7:     Let B_i be the quadrant containing the missing square x
8:     Place an L-shaped tromino at the centre so that it occupies one central square in
       each quadrant B_j, j ≠ i
9:     for j = 1, 2, 3, 4 do
10:        if j = i then
11:            x_j ← x
12:        else
13:            x_j ← the central square of B_j covered by the newly placed tromino
14:        end if
15:        TILE(B_j, n − 1, x_j)
16:    end for
17: end procedure
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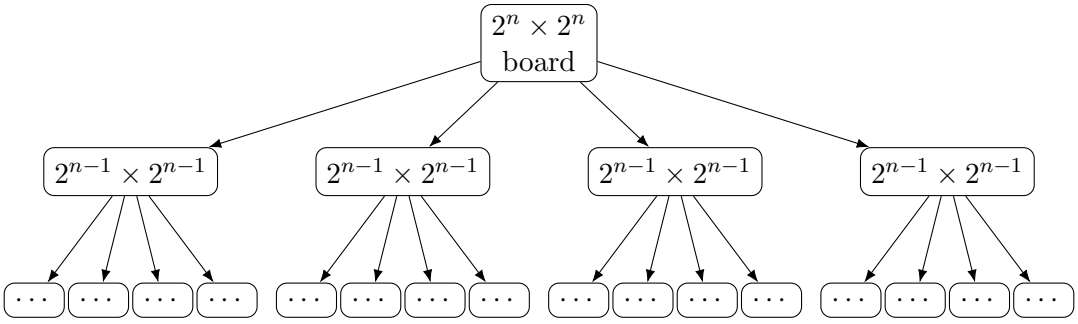


Figure 3: The recursive structure of the tromino tiling algorithm. Each problem is reduced to four problems with half the side length. The recursion eventually reaches 2×2 boards.

3 Induction and Recursion

Notice how closely the recursive algorithm follows the inductive proof. The two parts correspond almost exactly.

Inductive proof	**Recursive algorithm**
Establish the base case $P(1)$.	Directly tile a 2×2 board.
Assume that every smaller $2^{n-1} \times 2^{n-1}$ deficient board can be tiled.	Call TILE recursively on the four smaller deficient boards.
Use the induction hypothesis to prove the result for the larger $2^n \times 2^n$ board.	Use the four recursively constructed tilings, together with the central tromino, to construct the tiling of the larger board.

Mathematical induction and recursion are therefore closely related. Induction is a method for *proving* that a construction works for problems of every size, whereas recursion

is a method for *carrying out* the construction by solving smaller instances of the same problem.

There is an important difference in viewpoint:

Induction	:	If I can solve the smaller problem, I can prove that the larger problem can be solved;
Recursion	:	To solve the larger problem, actually solve the smaller problems using the same proced

The tromino problem makes this connection particularly visible.

4 A Recurrence for the Number of Trominoes

The recursive construction also allows us to count how many trominoes are used.

Let $T(n)$ denote the number of trominoes required to tile a $2^n \times 2^n$ board with one square removed.

For the base case,

$$T(1) = 1.$$

For $n > 1$, the algorithm first places one tromino at the centre. It then recursively tiles four boards of size $2^{n-1} \times 2^{n-1}$. Therefore

$$\boxed{T(n) = 4T(n-1) + 1.}$$

This is called a *recurrence relation*: it defines the value of $T(n)$ in terms of the value of the same function for a smaller argument.

We can also determine $T(n)$ simply by counting squares. A $2^n \times 2^n$ board contains

$$2^n\, 2^n = 4^n$$

unit squares. One square has been removed, so the number of squares that must be covered is

$$4^n - 1.$$

Since every tromino covers exactly three squares, the number of trominoes must be

$$T(n) = \frac{4^n - 1}{3}.$$

This agrees with the recurrence. Indeed, suppose

$$T(n-1) = \frac{4^{n-1} - 1}{3}.$$

Then

$$\begin{aligned} T(n) &= 4T(n-1) + 1 \\ &= 4\left(\frac{4^{n-1} - 1}{3}\right) + 1 \\ &= \frac{4^n - 4}{3} + 1 \\ &= \frac{4^n - 1}{3}. \end{aligned}$$

Thus

$$\boxed{T(n) = \frac{4^n - 1}{3}}.$$

For example, an 8×8 board corresponds to $n = 3$. It contains 64 squares, of which one is missing. Hence the tiling requires

$$T(3) = \frac{4^3 - 1}{3} = \frac{63}{3} = 21$$

trominoes.

5 Proof, Algorithm, and Recurrence

The tromino problem illustrates several ideas that recur throughout mathematics and computer science.

Starting with a single problem, we have obtained the chain

inductive proof $\longrightarrow$ constructive proof $\longrightarrow$ recursive algorithm $\longrightarrow$ recurrence relation.

The inductive proof tells us *why* a tiling must exist. The proof is constructive because it tells us how to build such a tiling. That construction becomes a recursive algorithm because it repeatedly reduces the problem to smaller instances of itself. Finally, examining the recursive algorithm naturally produces a recurrence relation.

This is an important connection between mathematical reasoning and computation: a proof may not merely tell us that a solution exists; sometimes it also contains an algorithm for finding one.